

Actors and use cases

Actors - warehouse operator, logistics supervisor, barcode/RFID scanner

Use cases

- Validate pallet placement
- Update pallet location
- Search for pallet
- scan pallet via barcode/RFID
- Monitor rack capacity

Use-Case Flow

Use Case

UC-02: Validate Pallet Placement

Primary Actor

Warehouse Operator

Main Success Scenario

1. The Warehouse Operator selects a pallet for storage.
2. The system retrieves the pallet's identification and weight information.
3. The Warehouse Operator enters or selects the proposed storage coordinates.
4. The system identifies the corresponding rack and location.
5. The system checks the current load and maximum load capacity of the selected rack.
6. The system calculates the rack load after the proposed pallet is added.
7. The system determines that the proposed placement does not exceed the rack's maximum load capacity.
8. The system confirms that the pallet placement is valid.
9. The Warehouse Operator receives confirmation that the pallet can be stored at the selected location.
10. The use case ends successfully.

Alternate Flow

7a. Rack Weight Capacity Exceeded (alternate from step 7)

1. The system determines that adding the pallet would exceed the rack's maximum load capacity.
2. The system rejects the proposed pallet placement.
3. The system displays a message indicating that the selected rack cannot safely accommodate the pallet.
4. The Warehouse Operator selects a different storage location.
5. The system repeats the validation process for the newly selected location.
6. The use case continues from Step 4 of the Main Success Scenario.